

Lec6

CSS Combinators

CSS Combinators are used to define the relationship between two or more selectors. They help select elements based on their position relative to other elements.

There are **4 main CSS Combinators**:

Combinator	Symbol	Meaning
Descendant	(space)	Selects all descendants
Child	>	Selects direct children only
Adjacent Sibling	+	Selects the immediate next sibling
General Sibling	~	Selects all following siblings

Descendant Combinator (Space)

Selects all matching elements inside a specified parent element, regardless of nesting level.

HTML

```
<div>
  <p>Paragraph 1</p>
  <section>
    <p>Paragraph 2</p>
  </section>
</div>
```

CSS

```
div p {
  color: blue;
}
```

Output

Both paragraphs become blue because both are descendants of `div`.

Child Combinator (>)

Selects only the direct children of an element.

HTML

```
<div>
  <p>Direct Child</p>

  <section>
    <p>Not Direct Child</p>
  </section>
</div>
```

CSS

```
div > p {
  color: red;
}
```

Output

Only **Direct Child** becomes red.

Adjacent Sibling Combinator (+)

Selects the immediate next sibling element.

HTML

```
<h1>Heading</h1>
<p>Paragraph 1</p>
<p>Paragraph 2</p>
```

CSS

```
h1 + p {
  color: green;
}
```

Output

Only **Paragraph 1** becomes green because it is immediately after **h1**.

General Sibling Combinator (~)

Selects all sibling elements that come after the specified element.

HTML

```
<h1>Heading</h1>
<p>Paragraph 1</p>
<p>Paragraph 2</p>
<p>Paragraph 3</p>
```

CSS

```
h1 ~ p {
  color: blue;
}
```

Output

All three paragraphs become blue because they are siblings that follow `h1`.

CSS Position

The **position** property in CSS is used to control how an element is positioned on a webpage.

There are **5 main position values**:

1. `static`
2. `relative`
3. `absolute`
4. `fixed`
5. `Sticky`

Interview Definition

The CSS **position** property specifies how an element is placed on a webpage. The five values are **static**, **relative**, **absolute**, **fixed**, and **sticky**, each controlling the element's position in a different way.

Static Position

Every HTML element has `position: static` by default.

Elements appear in the normal document flow.

`top`, `right`, `bottom`, and `left` do not work.

```
.box {  
  position: static;  
}
```

Example

```
<div class="box">Hello</div>
```


Relative Position

The element remains in the normal document flow.

You can move it relative to its original position using `top`, `right`, `bottom`, and `left`.

```
.box {  
  position: relative;  
  left: 50px;  
  top: 20px;  
}
```

Effect

The element moves 50px right and 20px down from its original position.

Absolute Position

The element is removed from the normal document flow.

Positioned relative to its nearest positioned ancestor (**relative**, **absolute**, **fixed**, or **sticky**).

If no positioned ancestor exists, it is positioned relative to the page (**body**).

```
.parent {  
  position: relative;  
}
```

```
.child {  
  position: absolute;  
  top: 20px;  
  left: 30px;  
}
```

Example

```
<div class="parent">  
  <div class="child">Box</div>  
</div>
```

The child is placed 20px from the top and 30px from the left of the parent.

Fixed Position

The element is removed from the normal flow.

Positioned relative to the browser window (viewport).

Stays in the same place even when the page scrolls.

```
.box {  
  position: fixed;  
  bottom: 20px;  
  right: 20px;  
}
```

Sticky Position

Behaves like `relative` initially.

Becomes `fixed` when a specified scroll position is reached.

```
.header {  
  position: sticky;  
  top: 0;  
}
```

Navigation Styling

Navigation Styling refers to the process of designing and formatting navigation menus (navbar) using CSS to make them visually attractive, user-friendly, and responsive.

A navigation bar typically contains links such as **Home, About, Services, Contact**, etc.

Basic Navigation Menu

```
<nav>
```

```
<ul>
```

```
<li><a href="#">Home</a></li>
```

```
<li><a href="#">About</a></li>
```

```
<li><a href="#">Services</a></li>
```

```
<li><a href="#">Contact</a></li>
```

```
</ul>
```

```
</nav>
```

CSS of menu

```
ul {  
list-style: none;  
margin: 0;  
padding: 0;  
background-color: #333;  
}  
li {  
display: inline-block;  
}  
a {  
display: block;  
color: white;  
text-decoration: none;  
padding: 15px 20px;  
}  
a:hover {  
background-color: #555;  
}
```

Common Navigation Styling Properties

Remove List Bullets

```
list-style: none;
```

Remove Default Margin and Padding

```
margin: 0;
```

```
padding: 0;
```

Display Menu Horizontally

```
li {  
    display: inline-block;  
}
```

Remove Underline from Links

```
a {  
    text-decoration: none;  
}
```


Layout

CSS Layout refers to the way elements are arranged and positioned on a web page. Layout techniques help organize content such as headers, navigation bars, sidebars, main content, and footers.

Interview Definition

CSS Layout is the technique used to arrange and position elements on a webpage. Common layout methods include Display, Float, Positioning, Flexbox, Grid, and Responsive Design, which help create structured and user-friendly web pages.

Display Property

The `display` property controls how elements appear.

Block Elements

Take the full width available.

```
div {  
  display: block;  
}
```

Examples:

```
<div>  
<p>  
<h1> to <h6>
```

Inline Elements

Take only the width they need.

```
span {  
  display: inline;  
}
```

Examples:

```
<span>  
<a>  
<strong>
```

Float Layout

Used to place elements side by side.

```
.box {  
  float: left;  
  width: 50%;  
}
```

Example

```
<div class="left">Left</div>  
<div class="right">Right</div>  
.left {  
  float: left;  
  width: 50%;  
}  
  
.right {  
  float: right;  
  width: 50%;  
}
```

Note: Float is an older layout method and is less common today.

Positioning

Controls element placement.

position: static;
position: relative;
position: absolute;
position: fixed;
position: sticky;

Example:

```
.box {  
  position: absolute;  
  top: 20px;  
  left: 50px;  
}
```

Homework

Try all positions

Task

Create a responsive navigation menu